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PROFESSIONAL SUMMARY

Diagnostics & Biosensing

Point-of-care diagnostics
Biosensor development
Optical sensing
Sensor optimization

Device & Assay Engineering

Microfluidics
Micro/Nanofabrication
Assay development
Surface Biofunctionalization

Technology Commercialization

NSF I-Corps Teams Award
Customer discovery
Translational research
Market validation

TECHNICAL SKILLS

- **Fabrication:** Electrochemical silicon etching, soft lithography, wax printing, SLA/extrusion 3D printing, paper microfluidics
- **Assay Development:** Immunoassays, PCR, surface biofunctionalization, paper-based diagnostics, 3D cell culture
- **Characterization:** SEM, TEM, Raman/SERS, UV-Vis, ^1H NMR, NanoSight
- **Software & Data Analysis:** Python, MATLAB, Origin, ImageJ/Fiji, AutoCAD, KLayout, Blender

RESEARCH & PROFESSIONAL EXPERIENCE

Vanderbilt University

Nashville, TN | Sep 2022 – Present

VU-Photonics Weiss Group & Locke Biosensing Lab

- **Diagnostic Platform & Microfluidic Design:** Developed a quantitative RDT platform integrating porous silicon optical sensors with paper microfluidics; engineered vertical- and lateral-flow architectures to control sample transport and assay incubation [2].
- **Sensor & Assay Development:** Engineered electrochemical silicon etching protocols to fabricate porous silicon sensors with a **6-fold faster response and 10-fold higher sensitivity** than conventional porous silicon biosensors; developed biotin–streptavidin and antibody-based immunoassays through surface biofunctionalization [3].
- **Cross-Lab Project Coordination:** Coordinated device fabrication, optical characterization, and assay validation across photonics and biosensing teams to meet project milestones.
- **Leadership & Mentoring:** Mentored students through the **VINSE Research Experience for Undergraduates (REU)** and **Vanderbilt Summer Academy Nanoscience and Nanotechnology** program.

Incheon National University

Incheon, Republic of Korea | Mar 2017 – Aug 2021

Biomaterial Engineering Intern (Song Lab) (Jul 2020 – Aug 2021)

- Synthesized stimuli-responsive hydrogels and engineered 3D-printed shape-deformable polymeric patches, optimizing multi-component formulations to control pH- and temperature-responsive transformation for localized drug delivery.

Bioelectronics Engineering Intern (BioElectronics & BioOptics Lab) (Mar 2019 – Jun 2019)

- Co-developed a vacuum-filtration fabrication protocol utilizing carbon nanotubes to construct conductive, transparent cellulose electronic devices [4].

AWARDS & COMMERCIALIZATION

- **NSF I-Corps Teams Award Recipient** | National Science Foundation | 2026
- Selected to evaluate the commercial potential of an optical biosensor through **customer discovery** and **market validation**.
- **Conference Parkhutik Prize Recipient** | 14th PSST International Conference | 2026
- Recognized for outstanding research on **fluid-flow control mechanisms** in paper-based porous silicon biosensors.
- **Pitch Competition Winner** | NSF Mid-South I-Corps Hub | 2025
- Awarded **first place and a microgrant** for an optical biosensor commercialization strategy.

EDUCATION

- **Vanderbilt University** | Nashville, TN | *Doctor of Philosophy (Ph.D.) in Interdisciplinary Material Science* | **Expected May 2027**
- **Incheon National University** | Incheon, Republic of Korea | *Bachelor of Engineering (B.E.) in Nano-Bio Engineering* | Feb 2021

SELECTED PUBLICATIONS & INTELLECTUAL PROPERTY

- [1] Kamat V., Xu K., **An H.**, et al., *Nano Letters*, 26(27), 8589-8599, 2026
- [2] **An H. et al.**, *Frontiers in Biological Detection: From Nanosensors to Systems XVII*, 13338, 16-22, 2025
- [3] **An H. et al.**, *ACS Applied Materials & Interfaces*, 17(4), 6024-6030, 2025
- [4] Noh S*, **An H***, et al., *Cellulose*, 28(5), 3081-3096, 2021
- **Patent Disclosure:** Porous silicon (PSi) and paper optical sensing device, Vanderbilt University, 2025